

BACKEND SOFTWARE DEVELOPMENT

BY

INFORMATION TECHNOLOGY



College of Computer Science and Information Technology

BELLS UNIVERSITY OF TECHNOLOGY-NEW HORIZONS

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May, 2025

NodeJS & NoSQL Database

(ICT 236)

Group 2

SUBMITTED TO

AYUBA MUHAMMAD

HOW WE CREATED IT

TOOLS USED:

- Gitbash
- VsCode
- MongoDB
- BlackBoxAI
- Node.js

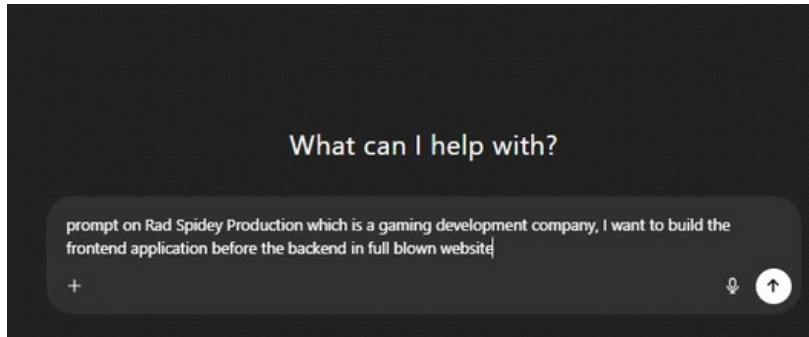
- Express
- CORS

GAME DEVELOPMENT COMPANY BACKEND

We Created the frontend and backend folders, Installed Blackbox AI extension into vscode and told it to "verify the folders in the project".

FRONTEND

After it verified the folders, i told chatgpt to give a "prompt on Rad Spidey Production which is a gaming development company, I want to build the frontend application before the backend in full blown website".



Copy & Paste the prompt to blackbox AI chat, it will start creating the code. And there we have our frontend code **"COMPLETED"**

BACKEND

For the backend, I told blackbox AI to generate the backend code for the frontend, meeting it with the requirements from the PDF.



it with Node.js, Express, MongoDB and CORS providing API endpoints for contact form submission.

- It also implemented additional backend features mentioned in the PDF such as user authentication, role based access control, API documentation, testing and deployment strategies.
- it added swagger API documentation setup in "Server.js" and created a swagger config.
- it created both an .env file and .gitignore file.
- it created the Readme.md

The Backend is now **"COMPLETE"**

It's been the AI doing the work now it's our turn,

- We created an account on mongoDB.

- To build



Log in to your account

Don't have an account? [Sign Up](#)

MongoDB 8.0 is here

Up to 32% higher throughput, improved horizontal scaling, expanded queryable encryption capabilities, and more.

- Created a Cluster and Chosed a cloud provider (AWS,West-3)
 - Created a database user. And finally it revealed the string

e.g **MONGO_URI=mongodb+srv://betinijames:ul3CmJtCKi7KdXCu@game-development-projec.909a1of.mongodb.net/**

This String will be used in the .env file blackbox AI created.

Paste it in the .env file below the port

e.g **Port = 5000**

After all that we'll open **gitbash**,

Add the backend directory

e.g

- **cd desktop/backend**

then install dependencies,

- **npm install**

then,

- **npm install bcrypt**

then,

- **npm install jsonwebtoken**

then,

```
Rad Spidey@DESKTOP-PP0UIFN MINGW64 /c/Bells University IT/ICT 235 Project (Mr.Mu
named)/2ND Semester/Game Development Production (FINAL)/Backend
$ npm start

> backend@1.0.0 start
> node server.js

Server running on port 5000
```

it should

show

> **backend@1.0.0 start**

> **node server.js**

server running on port 5000

MongoDB connected

NT: You must be connected to the internet,

This was successful because we implemented the mongoDB string to our .env file.

FRONTEND/BACKEND:

Back to frontend for the final result:

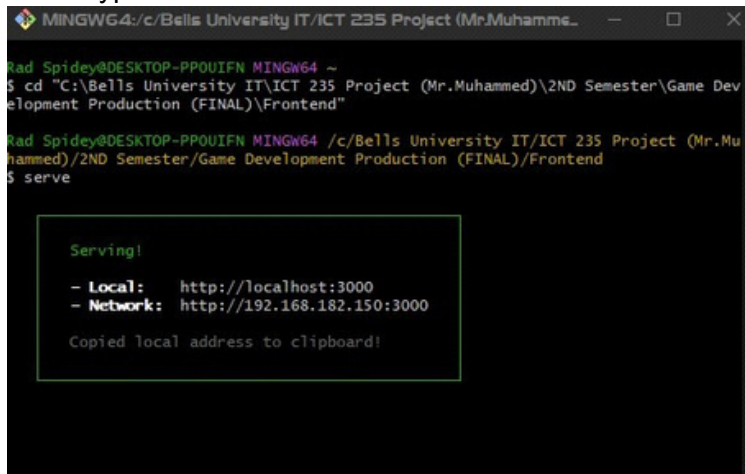
- Open gitbash
- install the serve package globally

npm install -g serve

then add the directory of the frontend

e.g **cd desktop/frontend**

- then type in **"serve"**

A terminal window titled 'MINGW64: c:/Bells University IT/ICT 235 Project (Mr.Muhamme...' shows the execution of the 'serve' command. The output indicates that the service is running on both local and network addresses. A green box highlights the output text: 'Serving!', '- Local: http://localhost:3000', '- Network: http://192.168.182.150:3000', and 'Copied local address to clipboard!'.

```
MINGW64: c:/Bells University IT/ICT 235 Project (Mr.Muhamme...
Rad Spidey@DESKTOP-PP0UIFN MINGW64 ~
$ cd "C:\Bells University IT\ICT 235 Project (Mr.Muhammed)\2ND Semester\Game Development Production (FINAL)\Frontend"
Rad Spidey@DESKTOP-PP0UIFN MINGW64 /c/Bells University IT/ICT 235 Project (Mr.Muhammed)\2ND Semester\Game Development Production (FINAL)\Frontend
$ serve

Serving!
- Local: http://localhost:3000
- Network: http://192.168.182.150:3000
Copied local address to clipboard!
```

This will give serve your frontend at a local URL shown in the terminal

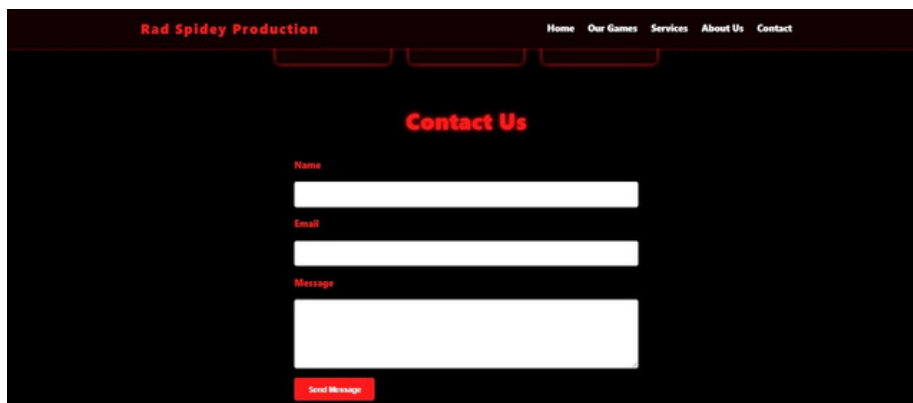
Local: <http://localhost:3000> or : <http://localhost:5000>

Copy this **local host** and paste it in your browser.

it will open the website that is implemented with both the frontend and backend.

Importantly,

Make sure the contact form submission works properly and shows all information in mongoDB database.



To show swagger API documentation,

- Go to the "Swagger.js"
 - Copy the URL and paste it in your browser
- e.g <http://localhost:5000/api-docs>